

HW Questions

Question 1: Time Complexity Analysis

Consider the following algorithm:

```
for i = 1 to n:  
  for j = 1 to n:  
    if A[i] == B[j]:  
      return True  
return False
```

a) What is the time complexity of this algorithm? b) Explain why this is the case. c) Provide an example where the worst-case scenario occurs.

Question 2: Time Complexity Analysis

Consider the following algorithm:

```
for i in range(n):  
    for j in range(i+1, n):  
        if A[i] == A[j]:  
            return True  
return False
```

- a) What is the time complexity of the algorithm?
- b) If $n=1,000,000$, estimate the number of comparisons performed in the worst case.

Question 2: Growth Rate Comparison

Given the growth rates of algorithms as illustrated in the chart comparing exponential, cubic, quadratic, linear, and logarithmic growth:

Rank the following complexities from fastest growing to slowest:

$O(n^2)$, $O(2^n)$, $O(n \log n)$, $O(\log n)$

Which complexity would be preferable for large input sizes and why?

